

PromptShala

Practical AI Literacy Studio for Educators

Version 1.3 · CRAFT decision and implementation alignment (8 Sep 2026)

Course	Educational Application Development — ET 617
Client	Centre for Educational Technology (CET), IIT Bombay
Project Leads	Prof. Ramkumar Rajendran & Aditya
Student Team	Darshan Sonawane · Vishal Patel · Ashok Kumar · Raghuram Gundi
Version / Date	Version 1.3 8 September 2026
Document Status	Draft (Revision 3) — incorporates CRAFT decision and implementation evidence

What changed through v1.3 (client feedback and 8 Sep implementation decision):

- Product name finalized as **PromptShala** — Practical AI Literacy Studio for Educators.
- Primary teacher persona expanded with classroom jobs, constraints, risks, behaviour and feature rationale.
- Official source repository recorded as github.com/EAD-Labs/Group-6_2026; GitHub and Jira working rules added.
- Milestones replaced with a **week-wise recovery and delivery schedule** following the 28 Aug review.
- Each delivery module retains separate staging, testing, evidence and client review.
- The complete project presentation remains fixed for **15 October 2026**.
- The prompt framework is finalized as **CRAFT — Context, Role, Action, Format and Target**.
- Privacy, factual verification and teacher review remain cross-cutting safeguards outside the five-letter structure.
- The signed Version 1.1 page is preserved unchanged as the final approval-record page.

1. Project Information

Project Title	PromptShala — Practical AI Literacy Studio for Educators
Client	Centre for Educational Technology (CET), IIT Bombay
Client Representatives	Prof. Ramkumar Rajendran and Aditya
Client Contact	ramkumar.rajendran@iitb.ac.in
Student Team	Darshan Sonawane (23b1541); Vishal Patel (23b0745); Ashok Kumar (23b0628); Raghuram Gundi (23b1534)
Primary Users	School and higher-education teachers, teacher educators, and professional-development participants
Application Type	Responsive web application; progressive-web-app compatible
Target Platform	Modern desktop, laptop, tablet and mobile web browsers
Official Repository	https://github.com/EAD-Labs/Group-6_2026 (created; initial project setup pending as of 2 Sep 2026)
HLD Version	1.3 (Revision 3)
Date / Status	8 September 2026 Draft for consolidated client review

1.1 Document Purpose

This HLD converts the client project brief into a development-ready design baseline. It defines users, workflows, architecture, data handling, scope, measurable acceptance criteria, responsibilities and the milestone agreement for the ET 617 student project.

Revision rule: Version 1.3 incorporates the client directions recorded on 21 and 28 Aug and the CRAFT decision recorded on 8 Sep. Any new scope or timeline change must be entered in the decision log and reflected in Jira. The original signed Version 1.1 page is retained unchanged as an approval-history record.

2. Project Overview

2.1 Problem Statement

Teacher professional-development programmes often introduce Generative AI through isolated workshops or demonstrations. Teachers may understand that AI tools exist but still lack sustained practice in structuring prompts, checking AI-generated output, creating reusable workflows, and applying AI responsibly to authentic classroom tasks. Without guided practice and formative feedback, adoption remains inconsistent and dependent on individual experimentation.

2.2 Proposed Solution

PromptShala is a self-paced, interactive professional-development platform that combines concise concept learning with guided experimentation. Participants progress through **four sequenced learning modules** covering AI Foundations and Responsible Use, Classroom Prompt Writing, Reusable Teacher Assistants, and Working with Teacher-Owned Sources. The platform provides quizzes, an AI-supported prompt sandbox, structured feedback, progress tracking, controlled module unlocking and a completion certificate.

Design principle: Learning is demonstrated through authentic teaching work — prompts, assistants and transformed resources — rather than passive content completion alone.

Note on terminology: the *learning modules* above are course content for participants. They are distinct from the *three delivery modules* used to build and test the software (see Sections 13 and 17.4).

2.3 Project Objectives

1. Build practical teacher competency in Generative-AI concepts, limitations, responsible use, APIs and prompting.
2. Develop participants' ability to create clear, contextual and testable prompts through repeated practice and formative feedback.
3. Enable teachers to design reusable AI assistants and repeatable workflows for lesson planning, assessment and resource creation.
4. Help teachers transform their own instructional materials into summaries, quizzes, study guides and related classroom resources while applying privacy and human-review safeguards.
5. Provide transparent progress, module-gating and completion evidence that supports individual learning and programme monitoring.

2.4 Deployment Context

Intended context (revised): Professional development for **school teachers and other educators**. PromptShala is delivered as self-paced online learning together with facilitated teacher-development workshops, and is intended for educators across schools and institutions — not for a single campus. CET / IIT Bombay is the commissioning client and pilot coordinator; the learning experience itself is aimed at school teachers and educators generally.

Expected users (revised): Proposed **initial pilot of 10 participants** (school teachers and educators). The architecture is designed for up to **25 concurrent authenticated users** to provide comfortable headroom during facilitated sessions and future cohorts.

Technology context: Participants may use personal or institutional laptops, tablets or phones over broadband or 4G. Core reading can tolerate intermittent connectivity; AI practice and authenticated progress updates require internet access.

3. User Personas

Persona note: The personas are fictional composites grounded in the client discussions of 7, 14, 21 and 28 Aug 2026. They describe design requirements, not research findings.

3.1 Primary Persona — Meera, Beginner School Teacher

Teaching context	Teacher of Classes 5–10, commonly Mathematics or Science; 5–15 years of experience; 35–45 learners per class; limited preparation time.
Current AI experience	Beginner. May have tried ChatGPT or Gemini for a question or worksheet, but does not consistently define role, learner, context, outcome or output format.
Digital context	Comfortable with WhatsApp, documents, presentations and web forms; uses a personal Android phone and sometimes a laptop; connectivity may vary.
Motivation	Save preparation time and create age-appropriate lessons, explanations, assessments, slides, parent communication and differentiated resources.
Main anxiety	Giving wrong information to learners, exposing student information, copying copyrighted content, or appearing unskilled when the tool produces poor output.
Time available	Short 20–30 minute learning sessions; needs immediate practice and visible improvement rather than long theory-first instruction.

Teacher job	Current difficulty	PromptShala support
Explain a concept	Generic AI explanations ignore grade, prior knowledge and misconceptions.	Scenario-based prompt practice requires learner level, teaching role, context and expected format.
Create assessments	Questions may be poorly levelled, repetitive or disconnected from learning outcomes.	Guided prompts cover outcome, Bloom level, question type, answer key and review checklist.
Prepare lessons/resources	Repeatedly rewrites similar instructions for worksheets, slides and summaries.	Gem-style assistant builder turns a validated prompt into a reusable teaching workflow.
Use own materials	Uncertain how to safely convert notes or a syllabus into useful resources.	NotebookLM walkthrough and content-transformation safeguards emphasise source grounding and no student PII.

Behaviour, Barriers and Design Response

Observed / Expected Behaviour	Design Requirement	Feature Evidence
Starts with a vague one-line request and accepts the first output.	Show what is missing without technical jargon and invite revision.	Visible rubric, dimension feedback and side-by-side attempt comparison.
Prefers doing a classroom task to reading AI theory.	Introduce AI at a high level through questions, examples and choices.	Interactive onboarding and authentic scenarios before optional readings/videos.
May be uncomfortable entering an API key.	Delay API setup until needed; explain purpose, local handling and non-storage clearly.	Guided API-key setup, consent message and provider-independent fallback.
Uses a phone and may lose connectivity.	Short screens, saved drafts, clear resume state and no horizontal scrolling.	Responsive UI, autosave, progress persistence and retry-safe error states.

Why This Persona Is Different from a Generic Professional User

Dimension	School Teacher	Banker / Architect / Generic Professional
Primary recipient	Children with varied age, prior knowledge and learning needs.	Adult clients, colleagues, regulators or technical stakeholders.
Typical output	Lesson plan, explanation, quiz, rubric, worksheet, parent message, classroom slides.	Financial analysis, compliance note, design specification, proposal or operational report.
Quality test	Age appropriateness, pedagogy, curriculum fit, accessibility, misconception handling and teacher verification.	Commercial, regulatory, technical or client-specific correctness.
Safety boundary	No identifiable student data; teacher remains responsible for every classroom use.	Domain confidentiality and professional controls, but not child-centred pedagogy.

Primary success criterion (20–30 minute experience): Meera can write and improve one classroom prompt using CRAFT — Context, Role, Action, Format and Target, explain what was missing, save a reusable Gem-style specification, identify when NotebookLM is useful, and state the privacy and human-review safeguards before using the output with learners.

3.2 Secondary Persona — Anil, Teacher Educator / Facilitator

Anil facilitates teacher-development programmes. He needs a consistent sequence, aggregate cohort progress, evidence of practice, common-misconception summaries and clear intervention points without routine access to private prompts or uploaded materials.

3.3 Supporting Roles

Persona	Primary Goal	Main Need
CET Content Curator	Maintain accurate, sequenced teacher-learning content.	Draft, preview, publish and version controls.
Programme Administrator	Operate cohorts and protect access.	User, role, certificate, audit and configuration controls.
Institutional Sponsor	Understand adoption and completion.	Aggregated, privacy-preserving programme reports.

4. User Roles & Permissions

The product uses role-based access control. 'Participant' refers to the teacher or teacher educator completing the course; the Student Team is a development team and is not an application user role.

Permission safeguard: Private prompts and uploaded materials are not exposed to facilitators or administrators for routine analytics. Any exceptional support access must be explicit, time-limited and audited.

Capability	Participant	Facilitator / Content Manager	Administrator
Authenticate and manage own profile	Own account	Own account	All accounts
View published learning content	Allowed	Allowed + preview drafts	Allowed + preview drafts
Complete quizzes and practice	Own attempts	Optional test mode	Optional test mode
Use prompt sandbox / AI tools	Own workspace	Own workspace + examples	Own workspace + policy controls
View progress	Own progress	Assigned cohort; aggregate by default	All cohorts; aggregate by default
Create or edit learning content	Not allowed	Draft and edit	Draft, edit and publish
Manage module prerequisites	Not allowed	Recommend	Configure
Manage users and roles	Not allowed	Not allowed	Allowed
Issue / revoke certificates	Download own	Recommend	Allowed with audit record
Export reports	Own completion summary	Assigned cohort	Programme-level
View raw private uploads	Own files only	Not allowed by default	Break-glass support only; audited

5. User Journeys & Core Workflows

5.1 Primary User Journey

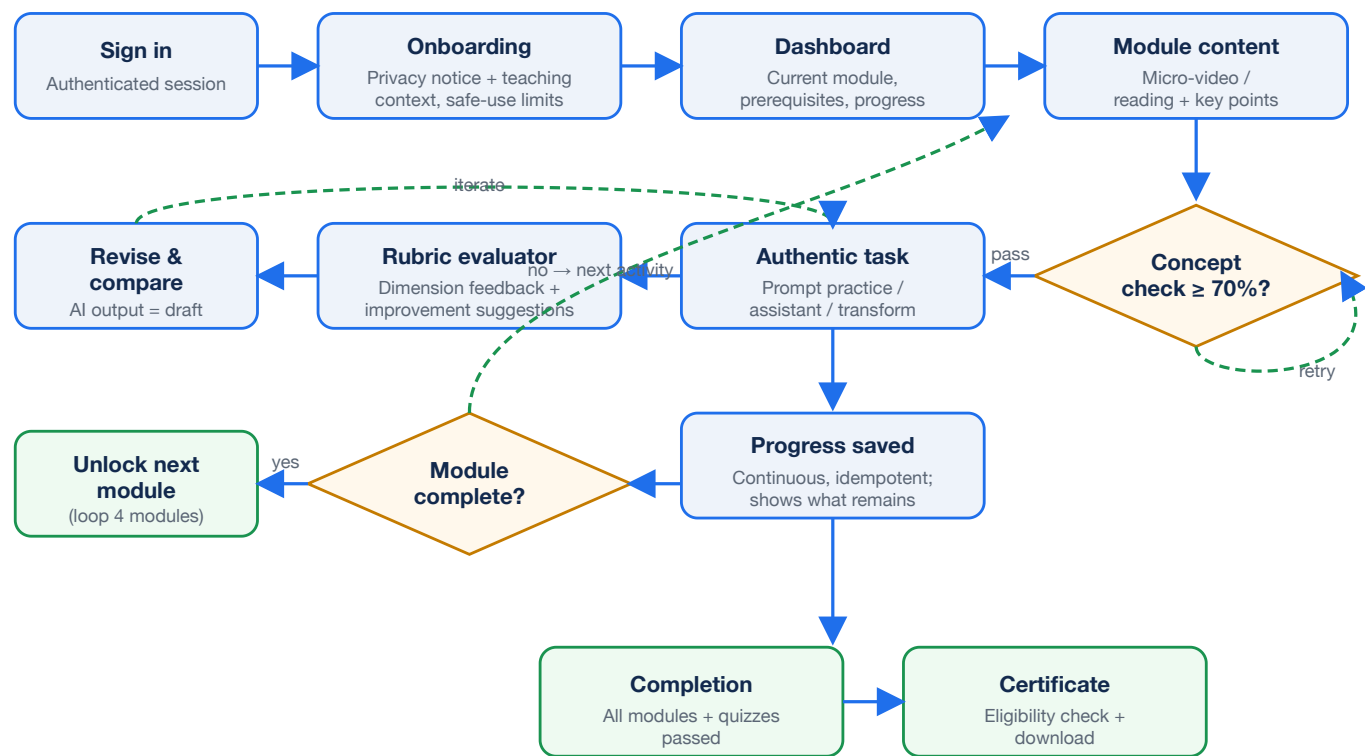


Figure 1. Primary participant journey — corrected to show the prompt-practice feedback loop, the quiz-gated module unlock ($\geq 70\%$), and completion $\rightarrow$ certificate.

A participant signs in, completes a short onboarding and sees a dashboard showing the current module, prerequisites and progress. Each module combines a micro-video or reading with a concept check and an authentic task. The system saves progress continuously and explains what remains before the next module unlocks.

During prompt practice, the participant submits a prompt to a rubric-driven evaluator. The platform returns dimension-level feedback, an explanation and improvement suggestions; the participant can revise and compare attempts. AI output is presented as a draft requiring teacher review, not as an authoritative classroom answer.

5.2 Secondary User Journey

1. Facilitator or content manager creates or revises a learning item in Draft status.
2. The item is previewed across desktop and mobile layouts and checked for accessibility.
3. An administrator approves publication and assigns the item to the appropriate module / cohort.
4. The facilitator monitors aggregated completion, quiz patterns and intervention flags.
5. The team revises content based on evidence while preserving historical attempt records.
6. At programme completion, authorized staff export a cohort summary and certificate register.

5.3 Key Use Cases

ID	User	Description	Expected Outcome
UC-01	Participant	Sign in and resume from the last completed activity	Authenticated session and accurate progress state
UC-02	Participant	Complete a lesson and auto-graded quiz	Immediate explanation, retry option and prerequisite update
UC-03	Participant	Submit and revise a prompt in the sandbox	Dimension scores, feedback and attempt comparison
UC-04	Participant	Configure a reusable teaching-assistant template	Saved assistant specification with purpose, context, constraints and output format

UC-05	Participant	Transform teacher-owned material	Editable draft summary, quiz or study guide with review checklist
UC-06	Facilitator	Review assigned-cohort progress	Aggregate completion and misconception indicators without exposing private content
UC-07	Administrator	Publish content and issue completion evidence	Versioned content and auditable certificate record

6. Functional Architecture

6.1 Major System Modules

ID	Module	Purpose and Main Functionality
M1	Identity & Access	Authentication, participant profile, roles, consent and secure session management
M2	Learning Experience	Four-module pathway, micro-videos, readings, examples and activity navigation
M3	Assessment & Progress	Quizzes, attempts, explanations, prerequisite rules, progress and completion state
M4	Prompt Practice	Guided sandbox, rubric evaluation, improvement suggestions and attempt comparison
M5	AI Workflow Tools	Reusable assistant specification, content transformation and NotebookLM workflow guide
M6	Certification	Eligibility checks, certificate generation, download and certificate register
M7	Administration & Insights	Content lifecycle, users / roles, cohorts, aggregate reports, configuration and audit review

6.2 Module Dependency

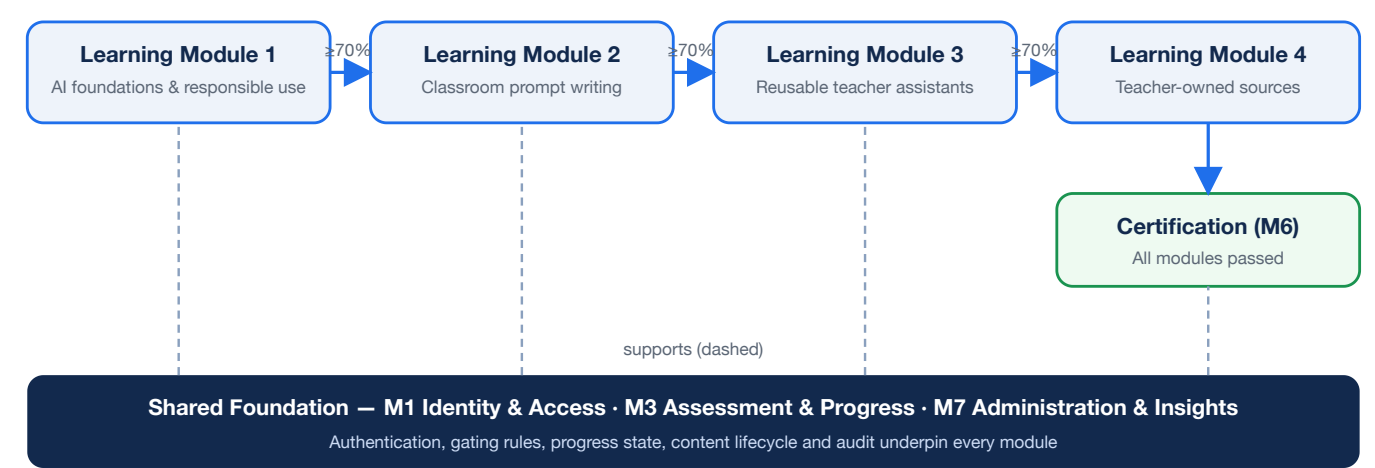


Figure 2. Functional module dependency and shared foundation.

Progression rule: A proposed module unlock occurs when all required activities in the preceding module are complete and the participant achieves at least 70% on its quiz. Retries are allowed and the highest valid score is retained.

7. High-Level System Architecture

7.1 Architecture Diagram

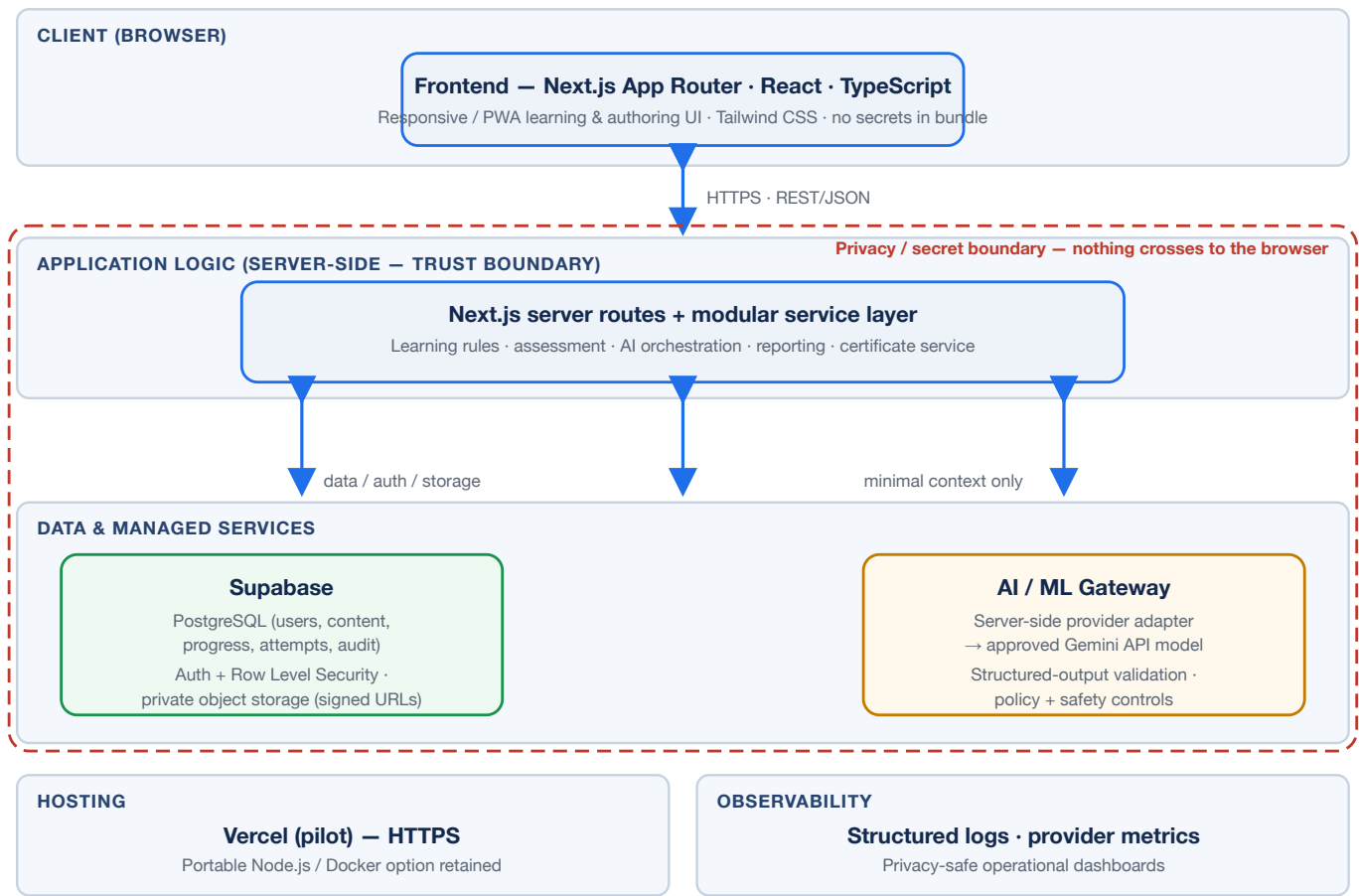


Figure 3. Portable, service-oriented web architecture — redrawn to show clear request flow, the server-side trust/privacy boundary, and separation of database, AI gateway, hosting and observability.

7.2 Architecture Components

Component	Purpose	Proposed Technology
Frontend	Responsive, accessible learning and authoring interface	Next.js App Router, React, TypeScript, Tailwind CSS
Backend / Application Logic	Learning rules, assessment, AI orchestration, reporting and certificate service	Next.js server routes and modular service layer; REST/JSON contracts
Database	Transactional users, content, progress, attempts, configuration and audit records	Managed PostgreSQL through Supabase
AI / ML Gateway	Prompt evaluation, suggestions and content transformation behind policy controls	Server-side provider adapter using an approved Gemini API model; structured output validation
Authentication	Email-based sign-in, sessions, roles and authorization	Supabase Auth with Row Level Security; optional IITB SSO in a future phase
Storage	Private uploaded materials, generated artifacts and certificate files	Supabase private object-storage buckets with signed access
Hosting	Pilot deployment, HTTPS, scaling and deployment automation	Vercel for the pilot; portable Node.js / Docker deployment retained as an institutional option
Observability	Error, latency, usage and security-event visibility	Structured application logs, provider metrics and privacy-safe operational dashboards

7.3 External Services

Service	Purpose	Dependency	Fallback / Control
Gemini API	AI feedback and content transformation	High for AI features	Rule-based rubric feedback, retry queue and clear service-unavailable state
Supabase	PostgreSQL, authentication and private storage	High	Documented backups and export; service boundaries isolated
Vercel	Pilot web hosting and deployment	Medium	Deployable as a standard Node.js / Docker application
GitHub	Official source control at EAD-Labs/Group-6_2026 , review and automated checks	High	Local repository mirror and release archives
NotebookLM	Guided external workflow only; no direct integration in MVP	Low	Static step-by-step guide and downloadable checklist

7.4 Architecture Principles

- **Secure by default:** authorization is enforced at both service and data layers, and secrets never enter the browser bundle.
- **Provider portability:** application logic depends on stable internal interfaces rather than vendor-specific calls throughout the codebase.
- **Data minimization:** store and transmit only information required for the learning task, progress evidence and approved operations.
- **Versioned evidence:** learning content, rubrics and certificate rules retain versions so historical attempts remain interpretable.
- **Graceful degradation:** non-AI learning remains usable during model-service disruption, while participant input is preserved for retry.
- **Observable and testable:** critical requests carry traceable IDs and the primary journeys are covered by automated and manual checks.

8. Data Architecture

8.1 Data Flow

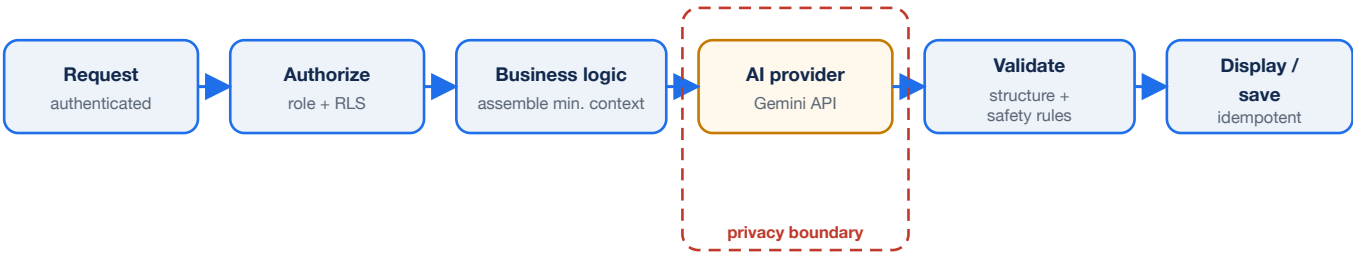


Figure 4. Learning and AI data flow with the privacy boundary around external AI processing.

Authenticated requests are authorized before business logic runs. AI requests are assembled server-side using the minimum required context. Provider responses are validated against expected structure and safety rules before being displayed or saved. Progress writes use idempotent operations so retries do not create duplicate attempts or certificates.

8.2 Key Data Entities

Entity	Description	Key Information
User / Profile	Authenticated participant or staff member	User ID, email, name, role, consent version, institution context
Cohort	Programme grouping	Cohort ID, dates, facilitator, status
Module / Lesson	Versioned learning structure	Title, order, prerequisite, content version, publish state
Quiz / Question	Concept assessment	Question, options, rationale, pass threshold, version
Attempt	Participant response record	User, activity, score, response, timestamps, completion state
Prompt Exercise	Authentic prompt task and rubric	Scenario, dimensions, criteria, required context
Prompt Submission	Participant prompt iteration	Prompt text, rubric result, feedback, revision link
Assistant Template	Reusable teaching-workflow specification	Purpose, audience, context, constraints, output format
Resource / Artifact	Teacher-owned upload or generated draft	Owner, type, storage key, expiry, transformation type
Progress	Derived pathway state	Completed activities, quiz status, unlocked module, completion percentage
Certificate	Completion evidence	Certificate ID, user, issue date, rule version, status
Audit Event	Administrative / security trace	Actor, action, resource, timestamp, outcome

8.3 Data Relationships

- A Cohort contains Users; a User has one active role per context and may enrol in multiple cohorts over time.
- A Module contains ordered Lessons and Activities; prerequisite rules determine availability.
- A User submits Attempts and Prompt Submissions; each record references the exact content or rubric version used.
- A User owns Assistant Templates, uploaded Resources and generated Artifacts; access is restricted by owner and role policy.
- Progress is derived from valid activity completion; Certificate issuance records the rule version that established eligibility.

8.4 Data Storage, Retention and Exclusions

Data Category	Proposed Storage / Retention	Control
Account and progress	Managed PostgreSQL; programme duration plus 12 months of inactivity	Export / delete workflow; client-configurable policy
Quiz and prompt practice history	PostgreSQL; aligned with progress retention	Owner access; aggregate analytics by default

Uploaded teaching materials	Private object storage; automatic deletion after 30 days or earlier user deletion	Signed URLs, type / size limits, no student PII
Generated artifacts	Private object storage; 30 days unless participant explicitly saves	Ownership checks and user-controlled deletion
Audit / security events	Restricted log store; 90 days proposed	Admin-only, tamper-evident export where feasible
Certificate register	PostgreSQL; three years proposed or client policy	Minimal fields; revocation status and audit trail

Data deliberately excluded: Student personally identifiable information, payment data, biometrics, raw passwords, client-side AI keys, and unnecessary demographic profiling. Participants are warned not to upload identifiable student work or confidential institutional material.

9. UI / UX Architecture

9.1 Screen Inventory

Screen	User	Purpose
Sign-in & Onboarding	All	Authenticate, accept privacy notice, set teaching context and explain safe-use boundaries
Learning Dashboard	Participant	Resume learning, view modules, requirements, progress and next recommended action
Module Overview	Participant	Understand outcomes, prerequisite status and activity sequence
Lesson Player	Participant	View micro-video / readings with transcript, key points and completion state
Concept Quiz	Participant	Answer questions, receive explanations and retry
Prompt Sandbox	Participant	Draft, evaluate, improve and compare prompts
Assistant Builder	Participant	Define and save a reusable teaching-assistant template
Content Transformer	Participant	Upload / paste material, choose output, review and export a draft
Progress & Certificate	Participant	Review evidence, incomplete requirements and download certificate
Content Studio	Facilitator / Admin	Draft, preview, version and publish programme content
Cohort Insights	Facilitator / Admin	Review aggregate progress, common errors and export summaries
Administration	Admin	Manage roles, cohorts, policies, prerequisites, certificates and audit events

9.2 Navigation Structure

Participant: Sign-in → Dashboard → Module → Lesson / Quiz / Practice → Feedback → Next Activity → Progress / Certificate. **Staff:** Sign-in → Content Studio or Cohort Insights → Draft / Preview / Publish or Review / Export. Navigation remains shallow, exposes one clear primary action per screen, and never relies on colour alone.

9.3 Conceptual Wireframes

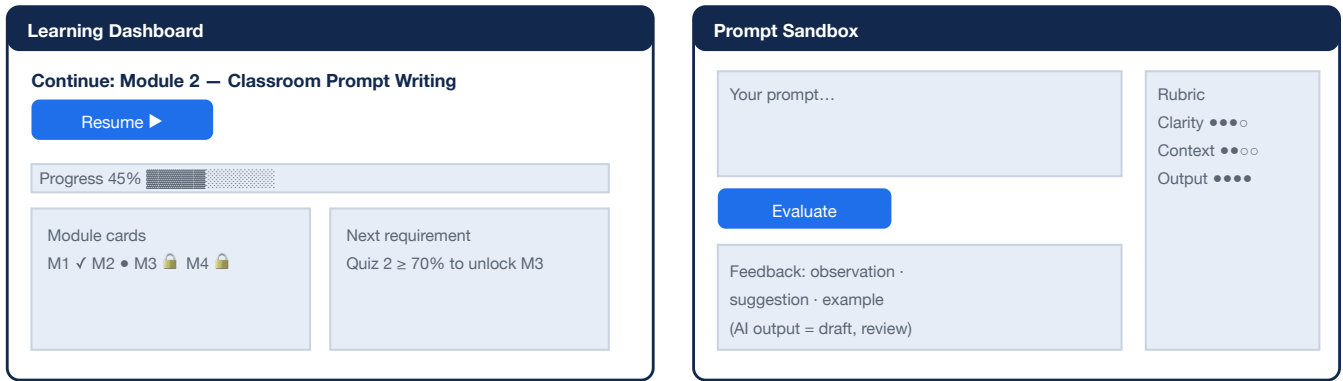


Figure 5. Conceptual dashboard and prompt-sandbox wireframes; visual design remains subject to client review.

Screen-Level UX Rules

- Every screen presents a visible title, current location and one dominant next action.
- Locked modules explain exactly what must be completed; disabled controls are not used without an explanation.
- AI feedback separates observation, suggestion and example so participants can decide how to revise.
- Long-running AI actions show progress, allow cancellation where feasible and preserve the participant's input on failure.
- Videos include captions / transcripts; quizzes and rubric results are keyboard and screen-reader operable.

10. Technology Stack

Status: The stack below is a proposed draft baseline chosen for type safety, rapid delivery, managed security controls and deployment portability. Final service accounts, regions and model selection require client approval.

10.1 Frontend

Item	Proposed Choice	Rationale
Platform	Responsive web / PWA-compatible	One codebase across laptops, tablets and phones; no app-store dependency
Framework	Next.js App Router with React	Server / client composition, routing and portable Node.js deployment
Language	TypeScript	End-to-end type safety and maintainable contracts
Styling	Tailwind CSS plus accessible component primitives	Consistent responsive system with controlled tokens
Client validation	Zod-compatible schemas	Shared, explicit validation for forms and structured AI output

10.2 Backend

Item	Proposed Choice
Framework / Runtime	Next.js server routes and modular TypeScript service layer on Node.js
API architecture	REST/JSON endpoints with versioned schemas; server-only AI and privileged data access
Authorization	Role policy plus PostgreSQL Row Level Security; deny by default
Background work	Idempotent jobs for AI retries, certificate generation and cleanup
Testing	Unit / component tests, API integration tests and browser-based end-to-end tests for critical journeys

10.3 Database and Storage

Database: Supabase-managed PostgreSQL for transactional and reporting data.

File storage: Private Supabase buckets; signed, short-lived access; lifecycle deletion for temporary materials.

Migration strategy: Version-controlled SQL migrations, seeded non-sensitive test data and documented backup / export procedure.

10.4 AI / ML Components

Item	Design
Model / Service	Approved Gemini API model accessed through a provider-neutral server adapter; exact model selected during implementation
Purpose	Prompt rubric feedback, revision suggestions and teacher-controlled content transformation
Inference	Cloud; no AI key or privileged prompt template is sent to the browser
Structured output	Schema-validated JSON for rubric dimensions, feedback and warnings before UI rendering
Safety	Provider safety settings, prompt-injection resistance, input limits, output checks and visible human-review reminder
Fallback	Deterministic rubric checklist and saved draft; retry option when the external AI service is unavailable

10.5 Infrastructure and Delivery

Area	Proposed Choice
Pilot hosting	Vercel with HTTPS and environment-managed secrets
Portable deployment	Standard Node.js server or Docker image for an approved institutional host
Version control	Official GitHub repository: github.com/EAD-Labs/Group-6_2026 ; protected main branch, feature branches and pull-request review

Work tracking	Jira project with epics for Recovery/Design, Delivery Modules 1–3, Integration/Release and Field Study; every weekly deliverable has an owner, due date, acceptance evidence and status
CI/CD	Automated lint, type-check, test, accessibility smoke check and deployment preview
Environments	Local, preview / staging and production with separate credentials and databases

10.6 Repository and Jira Working Agreement NEW

- **Repository of record:** https://github.com/EAD-Labs/Group-6_2026. The repository is owned by EAD-Labs so project history remains available after the course.
- **Contribution evidence:** all four students work through named accounts, feature branches and reviewed pull requests; direct commits to `main` are avoided.
- **Jira linkage:** branch names, commits and pull requests reference the Jira issue key. An issue is Done only when its acceptance evidence is linked.
- **Security:** API keys, tokens and production credentials are never committed; an `.env.example` documents required variables.
- **Weekly review:** Friday status uses the Jira board, staging URL, merged pull requests and test evidence rather than verbal progress alone.

11. Non-Functional Requirements

11.1 Performance and Reliability

NFR	Draft Target	Verification
Initial dashboard load	≤ 3 seconds on a representative 4G connection for cached / static assets	Browser performance test
Non-AI interaction	95th percentile ≤ 2 seconds under pilot load	API load test
AI feedback	Target ≤ 15 seconds; hard timeout ≤ 30 seconds with recoverable error	Instrumented integration test
Concurrent users	REVISED Up to 25 concurrent authenticated users for the 10-user pilot architecture (headroom for growth)	Staged load test
Availability	99.5% during scheduled pilot periods, excluding announced maintenance and external-service outage	Provider and application monitoring
Recovery	Documented daily backup / export and restore rehearsal before field study	Restore evidence

11.2 Accessibility

- Target WCAG 2.2 Level AA for participant-facing critical journeys.
- All functionality operable by keyboard with visible focus and logical focus order.
- Semantic headings, landmarks, labels, error association and meaningful alternative text.
- Captions and transcripts for instructional video; no essential information conveyed through colour alone.
- Minimum contrast and responsive text validated through automated checks and manual review.

11.3 Security

- TLS for all network communication; secure, HTTP-only session handling where applicable.
- Server-side secrets, least-privilege service roles and Row Level Security for user-owned data.
- Role checks at both API and database layers; administrative actions logged.
- Rate limits, request-size limits, input validation and safe error messages.
- Private file buckets, signed URLs, file-type / size validation and deletion lifecycle.
- Dependency, static-analysis and critical-flow security checks before production release.

11.4 Privacy and Responsible AI

Control	Requirement
PII collected	Name, email, role, institution / cohort context and learning records only as required
Student data	Not required and explicitly prohibited in uploads or prompts
Data minimization	Send only task-relevant context to the AI provider; do not include account metadata unless necessary
Transparency	Explain when AI is used, what is sent, limitations, retention and how the participant can delete material
Human oversight	All generated educational content is a draft; the teacher remains responsible for accuracy, bias, suitability and copyright review
Analytics	Aggregate by default; no public leaderboard or unnecessary comparison between participants
Third-party processing	Requires approved service configuration, contract / terms review and a client-approved privacy notice before pilot use

11.5 Compatibility

Supported devices: Current laptops / desktops, tablets and smartphones with a viewport of at least 360 CSS pixels.

Supported browsers: Current and previous major releases of Chrome, Edge, Firefox and Safari at release time; exact support matrix recorded during UAT.

11.6 Offline Requirements

The MVP is online-first. The application shell, previously opened text lessons and unsent draft input may be cached where safe. Authentication, AI evaluation, quiz submission, progress synchronization, administration and certificate issuance require connectivity. The interface must clearly indicate offline state and never imply that unsynchronized work has been submitted.

12. Assumptions & Constraints

12.1 Assumptions

1. The Client will supply or approve final learning content, micro-videos, quiz banks, prompt rubrics and certificate wording.
2. Participants have an email address, a modern browser and periodic access to stable internet; English is the MVP interface language.
3. **REVISED** The initial pilot includes **10 participants** (school teachers and educators) and does not require more than 25 concurrent authenticated sessions.
4. The Client will approve an AI provider, service account, usage budget, privacy notice and acceptable-use guidance before live AI features are enabled.
5. Participants use teacher-owned or non-sensitive materials and will not submit identifiable student information.
6. Completion requires all mandatory activities and at least 70% on each module quiz, with retries; this rule is subject to client confirmation.
7. The application supports professional learning and formative practice; it is not a high-stakes assessment, autonomous grading or decision-making system.

12.2 Constraints

1. The academic delivery window runs from August to November 2026 and the core team consists of four students.
2. External AI, hosting and managed-data services may impose availability, quota, pricing and policy constraints outside the Student Team's control.
3. AI output may be inaccurate, biased or unsuitable; safeguards reduce but cannot eliminate this risk.
4. Variable connectivity and device capability may affect video playback and AI response time.
5. Content, API access, approval and feedback delays may move dependent milestones.
6. Native mobile applications, enterprise integrations and multilingual content are outside the MVP unless scope is changed by mutual agreement.

13. Project Scope

13.1 In Scope — Minimum Viable Product

Scope Area	Included Capability
Access	Teacher authentication, secure sessions, onboarding, participant profile and role policy
Learning pathway	Four sequenced learning modules with micro-videos, readings, activity navigation and prerequisite-based unlocking
Assessment	Auto-graded concept quizzes, explanations, retries, completion thresholds and progress tracking
Prompt practice	Interactive sandbox, rubric-based AI evaluation, improvement suggestions and attempt comparison
Reusable AI workflows	Custom AI Assistant (Gem-style) specification builder with save / edit / export
Content transformation	Teacher-controlled transformation of material into summaries, quizzes and study guides with review checklist
NotebookLM	Guided workflow, examples and safety checklist; no direct API integration
Completion	Progress dashboard, eligibility calculation, completion certificate and certificate register
Administration	Content draft / preview / publish, module rules, cohorts, users / roles and aggregate reports
Quality	Responsive design, accessibility target, security / privacy safeguards, tests, deployment and documentation

13.2 Delivery Modules & Module-Level Testing NEW

Per client feedback, the build is split into **three delivery modules**. Each is developed, deployed to staging and **tested separately** (unit, integration and a module-level UAT / demo) before the next module begins. A per-module test report is shared with the Client at each module handover.

Delivery Module	System Modules Covered	Outcome Delivered & Tested Separately
Module 1 Foundation & Learning Core	M1 Identity & Access · M2 Learning Experience · M3 Assessment & Progress	Authenticated participants complete sequenced lessons and quizzes (retries), with prerequisite gating and progress state. Tested end-to-end before Module 2 starts.
Module 2 AI Practice & Workflows	M4 Prompt Practice · M5 AI Workflow Tools	Rubric-driven prompt sandbox, reusable assistant builder, content transformer and NotebookLM guide with safeguards. Tested separately, including AI-failure handling.
Module 3 Certification & Administration	M6 Certification · M7 Administration & Insights	Eligibility, certificate + register, content lifecycle, cohorts, roles, aggregate reports and audit. Tested separately.
Integration & Release		Cross-module end-to-end journeys, accessibility, performance, security and the complete project presentation — completed by 15 Oct 2026 .

13.3 Out of Scope for the MVP

- Native Android or iOS applications and full offline course completion.
- Learning-analytics warehouse, predictive / adaptive pathways or automated course recommendations.
- Personalized conversational AI coach beyond scoped feedback workflows.
- Teacher social network, peer prompt marketplace or collaborative communities.
- Institutional administrator dashboard across multiple organisations, SSO / LMS integration and institutional certificate management.
- Badge economy, gamification, multilingual interface and large-scale public deployment.
- Processing of identifiable student records, autonomous classroom decisions, high-stakes assessment or unsupervised publication of AI output.
- Long-term maintenance, paid production infrastructure and legal / compliance certification beyond the ET 617 project unless separately agreed.

Change control: Any functionality not explicitly listed as in scope requires a documented decision covering value, effort, privacy / safety impact, timeline and acceptance evidence before implementation.

14. Acceptance Criteria

The application is functionally complete when every Must-have criterion below is demonstrated in the approved deployment and no open Critical defect prevents a primary journey.

ID	Area	Acceptance Criterion	Evidence
AC-01	Access	A participant can authenticate, sign out and resume the correct account state; unauthorized role actions are blocked.	End-to-end test
AC-02	Learning	Four modules present approved videos / readings / activities in the agreed order and save completion state.	Content checklist + demo
AC-03	Quiz	Quizzes score automatically, show explanations, allow retry and retain a valid attempt history.	Test cases + demo
AC-04	Unlocking	A module unlocks only after required prior activities and the approved quiz threshold are complete.	Boundary tests
AC-05	Prompt sandbox	A participant can submit, receive dimension-level feedback, revise and compare at least two prompt attempts.	Scenario demo
AC-06	AI resilience	AI timeout or provider error preserves input and provides deterministic guidance plus retry without corrupting progress.	Failure-injection test
AC-07	Assistant builder	A participant can create, edit, save and export a reusable assistant specification.	Scenario demo
AC-08	Transformer	A participant can supply an allowed resource, select an output type, receive an editable draft and delete the resource.	Scenario demo
AC-09	Progress	Dashboard percentage and outstanding requirements match underlying activity records.	Data reconciliation
AC-10	Certificate	Eligible participants can download a certificate; ineligible participants see exact missing requirements.	Eligibility tests
AC-11	Administration	Authorized staff can draft, preview and publish content and configure cohort / module settings.	Role-based demo
AC-12	Reporting	Facilitator can view / export assigned-cohort aggregate completion without routine access to private prompts / uploads.	Permission test
AC-13	Responsive UX	Primary journeys work at 360 px mobile, tablet and desktop widths without clipping or horizontal page scroll.	Visual regression / UAT
AC-14	Accessibility	Critical journeys pass agreed automated checks and a manual keyboard / screen-reader smoke review against WCAG 2.2 AA target.	Audit report
AC-15	Performance	Measured response targets in Section 11.1 are met under the agreed pilot profile or deviations are accepted in writing.	Performance report
AC-16	Privacy	Consent, no-student-PII warning, private-file access and deletion / expiry behavior are implemented and documented.	Privacy checklist
AC-17	Security	No open Critical / High finding in the agreed pre-release security checklist; secrets are absent from client code / repository.	Security review
AC-18	Delivery	Production / staging deployment, source code, setup guide, user guide, HLD, test summary and presentation are handed over.	Handover checklist
AC-19	Module testing NEW	Each of the three delivery modules passes its own unit, integration and UAT / demo before the next module begins; a per-module test report is provided.	Per-module test reports

14.1 Defect Severity and Sign-off

- **Critical:** security / privacy breach, data loss or inability to complete a primary journey — must be resolved before release.
- **High:** major function fails with no acceptable workaround — must be resolved or explicitly waived by the Client.
- **Medium / Low:** limited impact or cosmetic issue — may be scheduled with an agreed owner and date.
- Client UAT feedback should be consolidated within each module's review window so the Student Team can assess and incorporate it before the next module.

15. Development Dependencies

Re-baselined on 2 Sep 2026: The client provides direction, review, approvals and suitable examples; the Student Team curates or creates the learning materials and implementation. Placeholder content may be used only until client confirmation.

Dependency / Decision	Owner	Required By	Status / Impact
Confirm PromptShala name, primary teacher persona and four learning outcomes	Client + Team	4 Sep 2026	Name selected by Team; persona/outcomes require consolidated client review
Curated or team-created sample content for AI introduction and prompt practice	Student Team	6 Sep 2026	Required for Module 1 prototype; Creative Commons or embedded YouTube only
Validate CRAFT terminology and example classroom scenarios	Client + Team	11 Sep 2026 review	CRAFT implementation baseline confirmed by Team on 8 Sep; client review validates wording and scenario priority
Privacy notice, acceptable-use wording and API-key explanation	Client + Team	10 Sep 2026	Required before any external participant use
AI implementation decision: lightweight rules/model first; teacher API key only when required	Client + Team	18 Sep 2026	Blocks production configuration for Module 2
Non-sensitive sample syllabus/worksheet for transformation testing	Client	25 Sep 2026	Required for Module 2 transformer test evidence
Hosting/domain credentials or approval to use team staging	Client	5 Oct 2026	Staging can continue independently until release
Pilot participant list and schedule (10 teachers/educators)	Client	10 Oct 2026	Required for field-study planning
Consolidated final UAT feedback and release decision	Client	13 Oct 2026	Required before final presentation on 15 Oct

16. Open Questions & Decision Log

Version 1.3 records the signed Version 1.1 baseline, the client directions from 21 and 28 Aug, and the CRAFT decision from 8 Sep. Items marked Confirmed are current working decisions; remaining items require consolidated client confirmation.

ID	Question / Decision	Proposed Decision	Owner	Date / Status
D-01	Delivery platform	Responsive web application; PWA-compatible; no native app in MVP	Client + Team	14 Aug — Proposed
D-02	End-user roles	Participant, Facilitator / Content Manager and Administrator	Client + Team	14 Aug — Proposed
D-03	Completion rule	Required activities plus $\geq 70\%$ on each module quiz; retries allowed	Client	14 Aug — Proposed
D-04	Technology baseline	Next.js / TypeScript, Supabase data services, server-side Gemini adapter, portable hosting	Team + Client	14 Aug — Proposed
D-05	Pilot capacity REVISED	Initial pilot of 10 participants (school teachers & educators); test for up to 25 concurrent authenticated users	Client	17 Aug — Proposed
D-06	Privacy boundary	No student PII; private uploads; 30-day temporary-file retention; aggregate analytics	Client	14 Aug — Proposed
D-07	Scope boundary	Client-brief MVP is in scope; listed future enhancements are out of scope	Client + Team	14 Aug — Proposed
D-08	Client confirmation	Prof. Ramkumar Rajendran and Aditya will both sign as Client representatives	Client	14 Aug — Confirmed by user
D-09	Delivery structure NEW	Project delivered as three separately built and separately tested modules; complete project presentation by 15 Oct 2026	Client + Team	17 Aug — Proposed
D-10	Deployment context NEW	Intended audience is school teachers and other educators (not a single campus); CET/IITB is client and pilot coordinator	Client + Team	17 Aug — Proposed
D-11	Product identity NEW	PromptShala — Practical AI Literacy Studio for Educators	Student Team	2 Sep — Confirmed
D-12	Repository of record NEW	EAD-Labs/Group-6_2026 ; Jira tracks all weekly work and evidence	Client + Team	2 Sep — Confirmed
D-13	Recovery schedule NEW	Re-baseline from 2 Sep with weekly evidence; separately test all three delivery modules; retain 15 Oct presentation	Client + Team	2 Sep — Proposed
D-14	Teacher-specific design NEW	Scenarios, rubrics, safety and outputs are designed for beginner school teachers, not generic professionals	Client + Team	28 Aug — Confirmed direction
D-15	Prompt framework NEW	CRAFT — Context, Role, Action, Format and Target; transparent rule-based checks first, with privacy and teacher review applied across every activity	Student Team	8 Sep — Confirmed

Approval action: Before signing, the Client should confirm the proposed items in D-01 through D-15 or record an alternative decision in this section. Any change must be reflected in the relevant HLD section and milestone plan.

17. Project & HLD Agreement

The Student Team and the Client confirm that they have reviewed and discussed the project requirements, proposed solution, user personas, workflows, system design, project scope, technical approach and deliverables described in this High-Level Design (HLD) Document. The HLD, together with this Agreement, forms the agreed basis for development of PromptShala within the ET 617 academic project.

17.1 Agreement on HLD

- The approved HLD represents the current agreed understanding of the project.
- The HLD defines intended users, user journeys, core functionality, system architecture, technology approach, data safeguards, scope, assumptions, constraints and acceptance criteria.
- The Student Team will use the approved HLD as the basis for design, development, testing and delivery.
- The Client will review the HLD and provide consolidated feedback or approval within the agreed schedule.
- A significant change to scope, privacy, functionality, acceptance criteria or external dependencies will be discussed before implementation.
- Approved changes may affect the milestone plan and will be incorporated only after mutual agreement and an updated decision record.

17.2 Agreement on Responsibilities

Student Team Responsibilities

- Complete agreed development work and demonstrate milestone deliverables.
- Deliver and separately test each of the three modules, providing a per-module test report.
- Maintain source code in the official EAD-Labs GitHub repository, track weekly work in Jira, and retain technical documentation, test evidence and the decision log.
- Apply reasonable security, privacy, accessibility and responsible-AI safeguards described in this HLD.
- Incorporate reasonable, consolidated feedback received within the agreed review period.
- Communicate technical risks, dependency issues or timeline impacts as soon as reasonably possible.
- Provide the final deliverables identified in Section 17.4.

Client Responsibilities

- Provide accurate content, non-sensitive sample data, service access, credentials and approvals required for development.
- Nominate an appropriate representative for meetings and milestone review.
- Review each module deliverable within the specified feedback period and provide clear, consolidated feedback.
- Approve the AI service, privacy notice, acceptable-use guidance and pilot conditions before live participant use.
- Communicate major requirement changes as early as possible and participate in scope / timeline decisions.

17.3 Timeline Dependencies

The development timeline depends on timely participation, content, access, approvals and feedback from both parties. If a required Client dependency is delayed, the corresponding module milestone may be revised. If the Student Team identifies a technical or implementation issue that materially affects scope, quality or timeline, it will communicate the issue promptly with proposed options.

17.4 Final Deliverables and Project Timeline

Final Deliverables

- **Executable Application:** final working educational application / prototype in the approved environment.
- **Source Code:** complete source in [EAD-Labs/Group-6_2026](#), configuration template, commit/PR history and release tag / archive.
- **Per-Module Test Reports:** separate unit / integration / UAT evidence for each of the three delivery modules.
- **User Documentation:** participant and administrator instructions for the delivered functions.
- **Technical Documentation:** HLD, architecture, data model, setup / deployment guide and test summary.
- **Project Presentation:** final demonstration and presentation of process, product and outcomes — **delivered by 15 Oct 2026**.
- **Field Study & Analysis:** approved study materials, data summary and analysis presentation, subject to course and client requirements.

Week-wise Recovery and Delivery Schedule REVISED

Status at re-baseline (2 Sep 2026): HLD revisions and preliminary concepts exist, but the official repository has no project code and a complete approved UI/UX prototype has not been demonstrated. The plan below is a recovery schedule; the 15 Oct presentation date remains fixed.

Week / Dates	Focus	Required Output	Evidence / Separate Test	Review
Recovery 2–6 Sep	Baseline, persona, design and setup	HLD v1.2; PromptShala identity; detailed teacher persona; finalized user flow/wireframes; Jira board; repository scaffold, README, branching rules and staging plan	Approved documents, Jira backlog, visible GitHub commits/PR, clickable design	6 Sep

			or complete wireframe walkthrough	
Week 1 7–13 Sep	Module 1A — Access & learning shell	Sign-in, onboarding, teacher profile, dashboard, four-module pathway, lesson/video/transcript shell and saved progress	Staging demo; access/role tests; responsive smoke test	13 Sep
Week 2 14–20 Sep	Module 1B — Assessment & progress	Auto-graded quiz, explanations, retries, 70% gate, unlock logic and progress calculation	Module 1 test report: unit, integration, end-to-end/UAT evidence	20 Sep; feedback by 21 Sep
Week 3 21–27 Sep	Module 2A — Prompt practice	Teacher scenarios, CRAFT rubric (Context, Role, Action, Format and Target), prompt submission, missing-dimension feedback, revision and attempt comparison	Rubric test cases; safe fallback; prompt-practice staging demo	27 Sep
Week 4 28 Sep–4 Oct	Module 2B — Reusable workflows	Gem-style assistant specification builder, save/edit/export, NotebookLM guide and safeguarded material transformation	Module 2 test report: AI failure, privacy and workflow tests	4 Oct; feedback by 5 Oct
Week 5 5–9 Oct	Module 3 — Certification & administration	Eligibility, certificate/register, content draft/preview/publish, roles, cohorts and aggregate progress	Module 3 test report: eligibility, RBAC, reporting and audit tests	9 Oct; feedback by 10 Oct
Week 6 10–13 Oct	Integration, QA and UAT	Cross-module journeys, defect fixes, accessibility, performance, security/privacy checks, user/admin guides and release candidate	Integrated test summary, UAT checklist, defect register and release decision	13 Oct
Release 14–15 Oct	Handover & presentation	Final application, source/release tag, setup guide, HLD, test reports, user docs and complete project presentation	Final handover checklist and live demonstration	15 Oct
Field Study 16 Oct–6 Nov	10-user pilot and analysis	Approved pilot, data collection, analysis and findings presentation	Study evidence and privacy-safe analysis	6 Nov

17.5 Revision and Signature Continuity

Signed baseline retained: The final page of signed Version 1.1 is appended unchanged as the last page of this revision, as requested by the Student Team. It records approval history for the earlier baseline. Version 1.3 changes are documented in the revision summary, decision log and week-wise schedule and remain subject to consolidated client review.

Module 2	6–19 Sep	Prompt sandbox, assistant builder, transformer, NotebookLM guide	Delivery Module 2 on staging + Module 2 test report	22 Sep
Module 3	20 Sep–3 Oct	Certification and administration / insights	Delivery Module 3 on staging + Module 3 test report	6 Oct
Integration & Release	4–15 Oct	End-to-end integration, accessibility, performance, security; complete project presentation	Final application, source, docs & presentation by 15 Oct	13 Oct (UAT)
Field Study	16 Oct–6 Nov	Pilot with 10 participants, data collection and analysis	Field-study & data-analysis presentation	6 Nov

17.5 Mutual Confirmation

Mutual confirmation: By signing below, both parties confirm that they have reviewed and discussed the project requirements and HLD, agree to use the approved HLD as the development baseline, and commit to timely collaboration, access, feedback and approvals necessary to pursue the agreed deliverables.

This Draft becomes the approved Version 1.1 baseline only when the approval decision, approval date and required signatures are completed. Digital-signature cells are intentionally blank. Each signer should insert one digital-signature block that includes the signing date in the same cell.

Digital Signatures — Student Team

Student Name (Roll No.)	Digital Signature & Date
Darshan Sonawane (23b1541)	Digital Signature Darshan Avinash Sonawane (23b1541) 17-Aug-26 09:37:33 PM
Vishal Patel (23b0745)	Digital Signature Vishal Chandra Patel (23b0745) 17-Aug-26 10:29:10 PM
Ashok Kumar (23b0628)	Digital Signature Ashok Kumar (23b0628) 18-Aug-26 12:57:46 AM
Raghuram Gundi (23b1534)	Digital Signature Gundi Raghuram (23b1534) 18-Aug-26 05:18:44 AM

Client Representatives

Client Representative	Organization	Digital Signature & Date
Prof. Ramkumar Rajendran	Centre for Educational Technology (CET), IIT Bombay	Digital Signature Ramkumar Rajendran (10001783) 18-Aug-26 10:23:15 AM
Aditya	Centre for Educational Technology (CET), IIT Bombay	

HLD Approval

Approval Field	Value
HLD Document Version	1.1 (Revision 1)
Document Status	Draft — pending approval
HLD Approval Decision	Approved / Approved with changes / Not approved
HLD Approval Date	_____